

AMD ACT II · TRACK 2

StyleCap

Grounded multi-style video captioning

One clip → four voices. Every caption locked to a verified visual fact sheet.

Formal · Sarcastic · Humorous-tech · Humorous-non-tech

Gemma 3 12B

AMD MI300X

ROCm

Fireworks AI

Docker

PROBLEM

Accuracy + style

10-minute container, LLM judge

- Judge scores factual accuracy and tone match per style
- Generic or hallucinated captions score near zero
- ~12 clips, 4 styles each, valid JSON required
- No API key injected at runtime — baked in Docker image

FOUR STYLES

Formal — objective, detail-rich

Sarcastic — dry, Ah yes... cadence

Humorous-tech — one dev metaphor

Humorous-non-tech — relatable one-liner

ARCHITECTURE

Fact sheet first

ffmpeg



Vision JSON



Write x2



Judge



results.json

- 6 uniform-seeking frames @ 768px per clip
- Parallel workers + 90s per-clip budget
- Degradation ladder — never empty output
- Schema repair guarantees valid JSON

GRADED MODELS

kimi-k2p6 — vision / grounding

glm-5p2 — write + batched judge

GEMMA ON AMD

MI300X end-to-end

Same pipeline, local vLLM

`google/gemma-3-12b-it`

Ground + write + judge — all Gemma via ROCm/vLLM

No Fireworks Gemma deploy — avoids \$7/hr credit drain

Demo video: [live rocm-smi + pipeline run](#)

BENCHMARK

86 seconds — 8 public clips

~40 GB VRAM (82% utilization)

Evidence: [GEMMA_RESULTS.md](#)

SUBMIT

Ready for grader

DOCKER IMAGE

ghcr.io/quantumbyte-01/amd-track2-captioner:v34

REPOSITORY

github.com/QuantumByte-01/amd-track2-captioner

linux/amd64 · Public GHCR · CANDIDATES=2 · 9-min soft deadline